ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** simplified representation	Simplification.
2	**(d)** any ratio	Variable.
3	**(d)** chemical reaction	Law.
4	**(a)** negative charge	Negative.
5	**(d)** scalar	Magnitude only.
6	**(d)** push or pull	Force.
7	**(c)** force x displacement	W = Fs.
8	**(a)** vibration	Vibration.
9	**(c)** cell	Cell.
10	**(a)** similar cells	Same function.
11	**(c)** variety of life	Diversity.
12	**(d)** making new individuals	Continuity.
13	**(a)** energy	Main source.
14	**(a)** well-tested explanation	Explains.
15	**(b)** filtration	Filter paper.
16	**(b)** cation	Cation.
17	**(c)** 2 electrons	2n squared.
18	**(c)** displacement	Area.
19	**(c)** more inertia	Mass.
20	**(d)** changes the direction of force	MA = 1.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Simplified representation of a real system (2).	2
22	Substance made of two or more components not chemically combined (2).	2
23	Mass is neither created nor destroyed in a chemical reaction (2).	2
24	Electron (-), proton (+) (1); Neutron (0) (1).	1+1
25	Total length of the path travelled (2).	2
26	A body remains at rest or in uniform motion unless a force acts (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Length: metre (1); Mass: kilogram (1); Time: second (1).	1+1+1
28	Clear vs turbid (1); Stable vs settles (1); Cannot be filtered vs can be filtered (1).	1+1+1
29	Matter has atoms (1); Atoms of an element are identical (1); Atoms combine in whole-number ratios (1).	1+1+1
30	Most alpha particles passed (1); Some were deflected (1); Few rebounded (1).	1+1+1
31	Scalar and vector (1); Distance is always positive (1); Displacement may be zero (1).	1+1+1
	OR: Dust from carpet (1); Passengers falling backward (1); Coin on a card (1).	1+1+1
32	Moving car (1); Flowing water (1); Flying bird (1).	1+1+1
	OR: Tuning fork (1); Drum (1); Vocal cords (1).	1+1+1

SECTION — D (Long Answer) — Answers — 6 × 5 = 30

Q.	Expected Answer / Value Points	Marks

33	Question (1); Hypothesis (1); Experiment (1); Observation (1); Conclusion (1).	1+1+1+1+1
34	Homogeneous (1); Heterogeneous (1); Solution examples (1); Suspension examples (1); Colloids (1).	1+1+1+1+1
35	Statement (1); Reactants and products (1); Burning of carbon (1); 12 g + 32 g = 44 g (1); Closed system (1).	1+1+1+1+1
36	Positive sphere (1); Electrons embedded (1); Plum pudding (1); Neutral atom (1); Drawback (1).	1+1+1+1+1
	OR: Distance (1); Displacement (1); Speed (1); Velocity (1); Acceleration (1).	1+1+1+1+1
37	First law (1); Second law (1); Third law (1); Examples (1); Formulas (1).	1+1+1+1+1
	OR: Force (1); Displacement (1); In the direction of force (1); Examples (1); Zero work cases (1).	1+1+1+1+1
38	Vibration (1); Medium (1); Compression (1); Rarefaction (1); Wave travels (1).	1+1+1+1+1
	OR: Membrane (1); Nucleus (1); Cytoplasm (1); Organelles (1); Functions (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** above 20 kHz	Higher.
2	**(c)** water	Through a membrane.
3	**(a)** epithelial tissue	Covering.
4	**(a)** Porifera	Pores.
5	**(c)** ovary	After fertilisation.
6	**(c)** much sunlight	High albedo.
7	**(d)** 2 km	Divide by 1000.
8	**(c)** mass of solute + mass of solvent	Sum.
9	**(c)** ionic	Transfer.
10	**(c)** atomic number	Z.
11	**(c)** 50 km/h	100/2.
12	**(a)** 2 m/s squared	F/m.
13	**(c)** 20 J	Fs.
14	**(b)** 100 oscillations per second	Definition.
15	**(b)** two identical cells	Growth.
16	**(a)** hinge joint	One plane.
17	**(c)** Reptilia	Scales.
18	**(c)** testes	Male.
19	**(b)** carbon dioxide	CO2.
20	**(c)** metre	m.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Force multiplied by displacement in the direction of the force (2).	2
22	A form of energy produced by vibrating bodies (2).	2
23	Basic structural and functional unit of life (2).	2
24	Group of similar cells performing a function (2).	2
25	Grouping of organisms based on similarities (2).	2
26	Production of new individuals of the same kind (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Cell wall (1); Plastids (1); Vacuole size (1).	1+1+1
28	Apical (1); Lateral (1); Intercalary (1).	1+1+1
29	Cell type (1); Body organisation (1); Nutrition (1).	1+1+1
30	Fission: Amoeba (1); Budding: Hydra (1); Spores: bread mould (1).	1+1+1
31	Fossil fuels (1); Deforestation (1); Agriculture and waste (1).	1+1+1
	OR: Burning (1); Rusting (1); Cooking food (1).	1+1+1
32	Solution = 200 g (1); 20/200 x 100 (1); 10 per cent (1).	1+1+1
	OR: Low melting point (1); Do not conduct (1); Soluble in organic solvents (1).	1+1+1

SECTION — D (Long Answer) — Answers — 6 × 5 = 30

Q.	Expected Answer / Value Points	Marks

33	Definition (1); Apical (1); Lateral (1); Intercalary (1); Features (1).	1+1+1+1+1
34	Monera (1); Protista (1); Fungi (1); Plantae (1); Animalia (1).	1+1+1+1+1
35	Definition (1); Fission (1); Budding (1); Spores (1); Vegetative (1).	1+1+1+1+1
36	Geosphere (1); Hydrosphere (1); Atmosphere (1); Biosphere (1); Interactions (1).	1+1+1+1+1
	OR: Physical: no new substance (1); Chemical: new substance (1); Examples of physical (1); Examples of chemical (1); Signs (1).	1+1+1+1+1
37	Homogeneous (1); Small particles (1); Stable (1); Transparent (1); No Tyndall effect (1).	1+1+1+1+1
	OR: NaCl (1); MgO (1); CaCl2 (1); Al2O3 (1); H2O (1).	1+1+1+1+1
38	Definition (1); Hydrogen (1); Chlorine (1); Uses (1); Same chemical properties (1).	1+1+1+1+1
	OR: Total distance 100 km (1); Total time 3 h (1); 33.3 km/h (1); Formula (1); Units (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** mixture of gases	Homogeneous mixture.
2	**(a)** Lavoisier	Lavoisier.
3	**(c)** positive charge	Positive.
4	**(d)** vector	Magnitude and direction.
5	**(a)** newton	N.
6	**(b)** joule	J.
7	**(d)** medium	To travel.
8	**(b)** Robert Hooke	1665.
9	**(b)** meristematic	Cell division.
10	**(d)** similarities and differences	Characters.
11	**(a)** binary fission	Two parts.
12	**(d)** atmosphere	Gases.
13	**(a)** a pattern	Regularity.
14	**(c)** evaporation	Water evaporates.
15	**(b)** anion	Anion.
16	**(b)** 8 electrons	2 x 4.
17	**(c)** acceleration	Slope.
18	**(b)** third law	Action-reaction.
19	**(c)** pivot of a lever	Fixed point.
20	**(d)** below 20 Hz	Lower.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Geosphere and hydrosphere (1); Atmosphere and biosphere (1).	1+1
22	Approximate calculation (2).	2
23	Filtration (1); Distillation (1).	1+1
24	A charged atom or group of atoms (2).	2
25	Combining capacity of an element (2).	2
26	Rate of change of velocity (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Be 2, 2 (1); B 2, 3 (1); C 2, 4 (1).	1+1+1
28	108 x 5/18 (1); = 30 (1); m/s (1).	1+1+1
29	F = ma (1); 1000 x 3 (1); 3000 N (1).	1+1+1
30	W = Fs (1); 20 x 5 (1); 100 J (1).	1+1+1
31	T = 1/f (1); 1/250 (1); 0.004 s (1).	1+1+1
	OR: Protects (1); Controls entry (1); Controls exit (1).	1+1+1
32	Transport (1); Defence (1); Clotting (1).	1+1+1
	OR: No vascular tissue (1); Need water (1); Small (1).	1+1+1

SECTION — D (Long Answer) — Answers — 6 × 5 = 30

Q.	Expected Answer / Value Points	Marks

33	Fuel burns (1); Gases expelled downward (1); Reaction pushes up (1); Third law (1); Space travel (1).	1+1+1+1+1
34	Ramp (1); Longer path (1); Less force (1); MA = l/h (1); Examples (1).	1+1+1+1+1
35	Audible (1); Infrasonic (1); Ultrasonic (1); Examples (1); Uses (1).	1+1+1+1+1
36	Process (1); Two cells (1); Growth (1); Repair (1); Same chromosomes (1).	1+1+1+1+1
	OR: Neuron (1); Dendrites (1); Axon (1); Impulse (1); Function (1).	1+1+1+1+1
37	Cotyledons (1); Roots (1); Leaves (1); Flowers (1); Examples (1).	1+1+1+1+1
	OR: Barrier (1); Chemical (1); IUD (1); Surgical (1); Uses (1).	1+1+1+1+1
38	Source (1); Speed (1); Distribution (1); Reflection (1); Use (1).	1+1+1+1+1
	OR: Atomic model (1); Globe (1); Ball and stick (1); Simplify reality (1); Change with evidence (1).	1+1+1+1+1

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